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|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | highest at the lowest position of the pendulum.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 17 | B | $T_1 = \frac{1}{480} \text{ s}, \text{ the number of cycles in } 0.25 \text{ s} = \frac{0.25}{1/480} = 120.0$ $T_2 = \frac{1}{482} \text{ s}, \text{ the number of cycles in } 0.25 \text{ s} = \frac{0.25}{1/482} = 120.5$ $T_1 = \frac{1}{480} \text{ s}, \text{ the number of cycles in } 0.75 \text{ s} = \frac{0.75}{1/480} = 360.0$ $T_2 = \frac{1}{482} \text{ s}, \text{ the number of cycles in } 0.75 \text{ s} = \frac{0.75}{1/482} = 361.5$ <p>At <math>t = 0.25 \text{ s}, 0.75 \text{ s}</math>, the difference between the two waves are in <math>(n + 0.5)</math> cycles, the two waves will arrive anti-phase and their resultant amplitude is zero.</p> |
| 18 | C | The voltage is maximum at anti-node.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |